

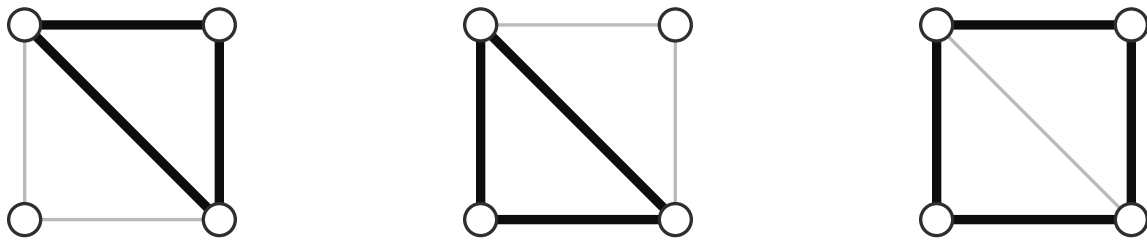
Fast Finisher / Bonus Challenges - adult key

Prepared September 30, 2026; **unpiloted reserve**. Student pages 1-4 contain Problems 1-4. Offer one page when a child wants another investigation; these are choices, not a speed test.

Entry and setup. Know the pair-flip rule and track several moves. Adults may read or scribe. Problems 1-3 need only small counts and picture matching; Problem 4 needs counting presses and adding costs up to 5. Use pencils/erasers, blank paper or the existing whiteboards; two-sided counters are optional. Print US Letter at Actual Size.

Launch and timing. Keep the usual common launch: let children try the materials, then show one line changing both lamps. At the bonus table say, 'Choose a question and show what one legal press does.' For Problem 3, demonstrate the special button separately. Allow about 10-15 minutes for 1 or 2, 10-20 for 3, and 5-10 for 4 within the existing hour. Staying with one problem is fine.

1. Do-nothing moves. A is impossible. B works by pressing the whole ring. C has exactly three answers, shown in bold below: either triangle, or the outside square. No other nonempty set works.



Reasoning. Each lamp must be changed an even number of times. On A, the endpoint forces its only line to be unused; continue inward. On C, an unused diagonal forces either no outside lines or all four. A used diagonal forces one of the two outside routes joining its ends. The successful sets all close up, touching each used lamp twice.

If needed: 'What happened to one lamp after all your presses?' **Extension:** add another line to a drawing and search for new do-nothing sets. Two closed loops meeting at a lamp show that a lamp may be changed four times too.

2. Bridge detective. A: **press** the bridge. Press the far-left vertical line and the bridge to reach the target. B: **leave it alone.** Press the far-left vertical line and the upper sloping line of the right triangle. Each is a two-press solution.

Reasoning. In A, all three lamps on the left must change, and only the bridge-end lamp on the right must change. In B, two lamps on each side must change. A line inside one side changes two lamps there; only the bridge changes just one. Thus odd change-counts force the bridge and even change-counts forbid it. The at-most-once rule makes this a yes/no choice; with repeats, it would mean odd/even numbers of bridge presses.

If needed: 'Which lamps differ between the two pictures?' **Extension:** invent a new pair of pictures with the same bridge prediction. Use changed lamps, not just the lamps ON in the target.

3. One solo switch. A: press the special button. B: press the button and a line to a neighboring lamp. C: press the button and the three consecutive lines leading to the opposite lamp. Every intermediate lamp changes twice, so only the far lamp remains ON. Either way around works; other choices of a far lamp also count.

D and the final question: yes, every target is possible. For any lamp other than the special one, press the button and every line along a route from the special lamp to it. The special lamp and all lamps in between change twice. Only the chosen lamp changes once. For the special lamp, use the button alone. Repeat for each desired ON lamp. Overlapping routes cause no problem: two identical presses cancel. An all-OFF target needs no presses.

Without the button, only even numbers of ON lamps could be made from all OFF. Now the odd targets work too. The six-lamp ring has 64 target pictures, all reachable. The same route argument works whenever every lamp can be reached from the special lamp along lines.

If needed: let the child light the special lamp, then ask how to move that light one place. **Extension:** move the special button elsewhere, or try a drawing with a separate piece. One button cannot remove the restriction on an unreachable piece.

4. Cheapest, not fewest. a: the top line, **1 press, cost 5**. b: the other three lines, **3 presses, total cost 3**. c: a press can have a different price from another press; taking more cheap steps can cost less than one expensive step.



Why these are best. Zero presses cannot change the start, so the one-press solution is shortest. If the top line is omitted, the two target lamps force the side lines; the two lower lamps then force the bottom line. If the top line is used, its endpoints force both side lines to be unused, and then the bottom line must be unused too. These are the only two sets of lines. Repeating a line twice changes nothing and increases the cost, so repetitions cannot produce a cheaper solution.

If needed: 'Can you avoid the line priced 5?' **Extension:** change its price to 3, then 2. At 3 both routes cost the same; at 2 the direct line is cheapest.

Sources and status. The four variants follow the organizer's September 30 brief. Visual format follows the two approved Week 2 catalogs. Earlier related work: Lowell Fall 2025, Handout 9, Problem 9.1 (divisor-based door switching). Format evidence: Natasha Rozhkovskaya, Math Circles for Elementary School Students, 'Introduction: Berkeley 2009' (adult help), Lesson 6, 'At the lesson,' item 4 (reported response to concrete geometry). These are teaching references, not sources for the new puzzle instances.

All finite instances were exhaustively checked. Record the exact bonus problem, child attempts, discoveries and explanations only after use. The layouts and pacing remain unpiloted. Detailed data, proofs and rebuild instructions: plans/week-02-bonus.md.